

Rishu Kumar Singh

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SKILLS

Technical (Core System & Low-Level): NASM (x86 – 16/32/64-bit), Compiler Design, Operating Systems
Languages: C, C++, Java, Python, PHP, JavaScript
Frameworks: HTML and CSS, Bootstrap, NodeJS, React
Database: MySQL, MongoDB, B-Tree Indexing, Query Optimization, Red-Black Tree, B-Tree, B+ Tree
Tools/Platforms: GitHub, Git, Linux, Agile (Scrum), Version Control, Docker, AWS, GDB
Soft Skills: Team Collaboration, Critical Thinking, Problem Solving, Adaptability, Time Management

PROJECTS

- Custom Programming Language & Compiler | [GitHub](#)**

Jul' 2025 - Present

 - Architected a **custom programming language** with full compiler support in **C++**
 - Developed core compiler components **Lexer, Parser, Semantic Analyzer, AST, and Assembly Code Generator**
 - Produced **optimized NASM assembly** with **~25% faster compilation** compared to baseline

Tech: C++, NASM, Compiler Design
- Database Engine | [GitHub](#)**

Jun' 2025

 - Engineered a **complete custom database engine** in C++ capable of handling large-scale read/write operations efficiently
 - Constructed the full backend architecture including **Lexer, Parser, AST generator, Semantic Analyzer, and B-Tree indexing module** from scratch
 - Tuned execution pipeline and memory management, enabling the system to sustain **1000+ concurrent queries/second with consistently low-latency response**

Tech: C++, Crow Framework, B-Tree Index File Storage
- AI Debugger Assistant | [GitHub](#)**

Apr' 2025

 - Created an **AI-powered debugger and code analyzer** using **OpenAI API**
 - Designed a responsive UI using **HTML, Tailwind CSS, JavaScript**
 - Enabled real-time optimization and bug-pattern detection, **reducing debugging time by ~30%**

Tech: OpenAI API, JavaScript, Tailwind CSS, Debugging Automation

TRAINING

- Mastering Data Structures & Algorithms | LPU Skill Development | [Certificate-Link](#)**

Jul' 2025

 - Applied core DSA concepts to design efficient data handling and optimization strategies
 - Built an **Interactive Red-Black Tree Visualizer** with real-time rotations and auto-balancing
 - Enhanced proficiency in balanced tree operations, rotations, and **O(log n)** implementations

CERTIFICATES

- Oracle Cloud Infrastructure DevOps Professional | [Oracle](#)

Oct' 2025
- Introduction to Hardware and Operating Systems | [Coursera](#)

Oct' 2024
- The Bits and Bytes of Computer | [Coursera](#)

Sep' 2024
- Ethical Hacking | [Rising Tech Pro](#)

Feb' 2024

ACHIEVEMENTS

- Solved 200+ Data Structures & Algorithms problems on LeetCode**

Dec' 2025

EDUCATION

- Lovely Professional University**

Punjab, India

Bachelor of Technology - Computer Science and Engineering; **CGPA: 8.22**

Aug' 2023 - Present
- Sachchidanand Sinha College**

Aurangabad, Bihar

Intermediate; **Percentage: 72.6%**

Apr' 2021 - Mar' 2022
- Saraswati Shishu Mandir**

Aurangabad, Bihar

Matriculation; **Percentage: 85.4%**

Apr' 2019 - Mar' 2020